Measuring the Coordination-Collapse Boundary in Policy-Gradient Cooperative MARL

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Abstract

Cooperative multi-agent reinforcement learning (MARL) can collapse into a degenerate *lazy-agent* policy when reward centralization is too low, but the field describes this failure only qualitatively. Our contribution is a way to measure it: a pre-registered collapse definition and bootstrap confidence intervals that turn the failure into a critical centralization w_{crit} with error bars. We then use the instrument to ask whether behavioral preferences move the threshold, and find that they do: *loss aversion* systematically shifts w_{crit} rightward, so loss-averse agents need more centralization to coordinate. This effect is monotone in the loss-aversion strength λ and replicates in direction across four environment-algorithm combinations; at the human range ($\lambda \approx 3$) it adds $+0.089$ to the required centralization net of curvature ($+0.188$ in isolation). We report what is robust (the boundary, and the direction of the loss-aversion effect) separately from what is substrate-specific (the effect's magnitude and functional form), and offer a supporting mean-field account at the level of sign, not magnitude. The pre-registered protocol is a reusable instrument for measuring where any cooperative system collapses, and how far behavioral bias moves that point.

Introduction

A team of learning agents sharing a task can fall into a trap their designers never intended: instead of coordinating, every agent quietly defaults to inaction, and the team settles into a stable, useless equilibrium: *coordination collapse*, whose endpoint is the *lazy-agent* policy (Liu et al. 2023; Shi and Peng 2023), an instance of the sequential social dilemma that arises when private incentives drift from the collective good (Leibo et al. 2017). Whether a team falls in is governed by one design dial, the *reward centralization* $w \in [0, 1]$: the share of each agent's reward tied to the team's outcome rather than its own. Turn w down and an agent's private performance drowns out its contribution, so cooperating earns no learnable signal and the safe policy wins. The field knows this only as a direction (*more centralization helps* (Akella 2025; Liu et al. 2023)) and stops there. *Where*, in w, a working team tips into collapse has never been measured. Our central question is therefore twofold: does coordination collapse admit a *measurable* critical centralization threshold, and do behavioral preferences systematically shift it? The first half is a measurement problem (the paper's primary contribution), and the second is what the measurement lets us ask. The nearest work mitigates the lazy-agent pathology without locating a boundary (Liu et al. 2023; Shi and Peng 2023; Weng and Lee 2026), or sweeps neighbouring axes such as credit-assignment level (Zhao and Xie 2025) and reward weights (Jiang et al. 2020). Closest in spirit, Yamaguchi (2025) build a phase diagram for independent Q-learning over *environmental* axes (grid size, density), and Capannoli et al. (2026) prove a metastable boundary in network opinion dynamics under a fixed update rule; neither measures a reward-structure threshold in gradient-trained cooperative policies, with confidence intervals, that a behavioral bias then moves.

Pinning that number down matters now, because deployed agents are no longer neutral optimizers. In automated replenishment, fraud review, and energy dispatch, learning agents absorb human preference structure by construction: trained on human choices through preference-aligned objectives that are themselves prospect-theoretic (Ethayarajh et al. 2024), or sharing the loop with human operators (Lalmohammed et al. 2025). The sturdiest such regularity is *loss aversion*: a loss looms larger than an equal gain (Kahneman and Tversky 1979; Tversky and Kahneman 1992). Behavioral economics knows what this does to one chooser but is silent on a *group* learning to coordinate (Leibo et al. 2017; Lowe et al. 2017). A measured collapse boundary is the instrument that question needs, extending a thin existing bridge between behavioral preferences and collective learning dynamics (Hughes et al. 2018; Barfuss, Donges, and Kurths 2019; Hu, Leung, and Leung 2019) to the prospect-theoretic preferences today's training pipelines instill.

We therefore build the ruler first, and treat it as the paper's central object. Following the posture physics

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takes toward a phase boundary, we fix a binary collapse criterion in advance, freeze it, fit $P(\text{collapse} \mid w)$, and bootstrap confidence intervals on w_{crit}. ("Phase boundary" is used in the empirical, learning-dynamics sense of Yamaguchi (2025); Capannoli et al. (2026), not as a thermodynamic-criticality claim.) Two design choices keep the readings honest: we apply cumulative prospect theory (CPT) as a *reward transform* after blending rather than inside the update, so the optimizer is bit-identical across conditions and any movement of w_{crit} is behavioral; and we re-measure at several horizons, since a drifting threshold could pass for a constant. Our *primary* contribution is (i) a confidence-intervalled measurement of the coordination-collapse boundary in deep cooperative MARL, with a reusable pre-registered protocol. Using that instrument, we then establish three findings about behavioral bias: (ii) loss aversion shifts the boundary rightward, a direction whose sign we fix a priori and confirm across four environment-algorithm combinations, with the metastable horizon slide generalizing likewise; (iii) prospect theory's two components move it in *opposite* directions; and (iv) on a directed-flow substrate, a downstream loss-averse *minority* can set the boundary for the whole system. Our novelty is narrow and stated against the closest prior work below: what is new here is the measured, CI'd boundary and the component decomposition against it, in gradient-trained neural policies.

Related Work

Coordination collapse and credit assignment. Cooperative MARL, studied since independent Q-learning on shared tasks (Tan 1993; Buşoniu, Babuška, and De Schutter 2008), inherits the lazy-agent and relative-overgeneralization failures in which agents abandon cooperation for a degenerate policy (Liu et al. 2023; Shi and Peng 2023; Tampuu et al. 2017). The standard remedy is centralized credit assignment under CTDE, a centralized critic or factorized value function (Lowe et al. 2017; Yu et al. 2022; Sunehag et al. 2018; Rashid et al. 2018; Foerster et al. 2018; Son et al. 2019), benchmarked on suites like SMAC (Samvelyan et al. 2019); related work ties the failure to decentralized exploration (Weng and Lee 2026) or shows decentralized shaping cannot match centralized baselines (Akella 2025). This establishes *that* centralization helps and *how* to implement it; none locates the threshold w_{crit} or gives it a confidence interval, the gap we close, using w as the dial rather than a fixed factorization.

Phase transitions in multi-agent learning. Treating collective learning as a dynamical system with critical points has precedent: Yamaguchi (2025) build a phase diagram for independent Q-learning over *environmental* axes, Capannoli et al. (2026) prove a metastable boundary in opinion dynamics under a fixed update rule, and the mean-field programme for multi-agent learning (Barfuss, Donges, and Kurths 2019; Hu, Leung, and Leung 2019; Hussain, Belardinelli, and Leonardos 2025) derives Fokker–Planck descriptions of networked Q-learning. We share this statistical-physics posture but differ in object, ours is a *reward-structure* threshold with confidence intervals that a *behavioral* parameter moves, and use the mean-field machinery only to fix a sign.

Behavioral preferences in cooperative AI. A thin bridge connects human preference structure to collective learning dynamics: Hughes et al. (2018) show inequity aversion improves cooperation within the cooperative-AI agenda (Dafoe et al. 2020; Hammond et al. 2025; Barfuss et al. 2020). Risk-sensitive RL optimizes tail outcomes (Qiu et al. 2021; Shen et al. 2023; Prashanth 2014; Tamar, Glassner, and Mannor 2015), distinct from loss aversion, a reference-point asymmetry, and prospect theory has entered single-agent RL with known calibration concerns (Prashanth et al. 2016; Rabin 2000; Tversky and Kahneman 1992). In the multi-agent setting, Mengesha, Sun, and Roy (2025) prove loss aversion shifts a tipping point in evolutionary game theory over fixed strategies, and Lalmohammed et al. (2025) bring CPT into cooperative MARL but fold it into the update, entangling its components. We differ at the level of *apparatus*: rather than proposing another behavioral model, we build a measured boundary and use it as a common instrument, the behavioral study is its first application, not the contribution.

Preliminaries and Method

In cooperative MARL, N agents share an environment and maximize a joint objective under CTDE (Lowe et al. 2017); the reward-mixing weight $w \in [0,1]$ interpolates purely local (w=0) and purely global (w=1) reward. As $w \to 0$ the cooperative signal is diluted and agents collapse to the safe lazy policy (Liu et al. 2023; Shi and Peng 2023). This w is our sole structural intervention; all else is held fixed, so any boundary shift is attributable to w and the behavioral parameters. Cumulative prospect theory (Kahneman and Tversky 1979; Tversky and Kahneman 1992) gives the value function $v(x) = x^{\alpha}$ for $x \geq 0$ and $-\lambda(-x)^{\beta}$ for $x < 0$, where $\alpha, \beta \in (0,1]$ govern curvature and $\lambda > 1$ is loss aversion (α=β=1, λ=1 is linear utility). We apply it as a reward transform after blending,

$$r_i^{\text{PT}} = v\big((1-w)\, r_i^{\text{local}} + w\, r^{\text{global}}\big)\,, \tag{1}$$

the decisive difference from Lalmohammed et al. (2025), who fold CPT *into* the actor–critic update, entangling curvature with loss aversion and confounding any shift with a change to the optimizer. Transforming only the scalar reward leaves the policy gradient identical across conditions, so a measured shift is behavioral and the two channels dial independently. We pre-registered the binary collapse label $\mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1]$ ($\bar{S}$ mean base-stock, $\bar{\pi}$ mean profit), freezing it in version control before any analysis; the frozen label and its analysis code ship in the anonymized supplementary code bundle so the pre-registration is verifiable at review

time, with the de-anonymized commit hash and public repository released on acceptance. It is robust to ±20% threshold perturbation (max $\Delta w_{\text{crit}} = 0.015$; Supplement). For each condition we fit $P(\text{collapse} \mid w) = \sigma(b_0 + b_1 w)$, estimate $w_{\text{crit}} = -b_0/b_1$ (the w at which $P(\text{collapse}) = 0.5$), and bootstrap its 95% CI; steep high-λ cells use Firth penalized regression, whose w_{crit} agrees with the bootstrap estimate to within 0.019 at every λ, two independent estimators concur, so the boundary locations are not an artifact of either fitting method. The logistic midpoint is the natural transition point: collapse is a *binary* per-run outcome, so the sigmoid is its maximum-entropy model and $-b_0/b_1$ is the location parameter (the w of maximal uncertainty). The transition is empirically *sharp*, collapse fractions rise from near-zero to near-one within $\Delta w < 0.3$, so this is the center of a narrow, well-identified crossing, not an arbitrary cut through a gentle slope; we report its width Δw alongside its location. We use "phase boundary" in this empirical sense, not as a thermodynamic-criticality claim. The decomposition compares three conditions with identical environment and optimizer: **Null** ($\alpha{=}\beta{=}1, \lambda{=}1$), **Curvature** ($\alpha{=}\beta{=}0.88, \lambda{=}1$), and **Loss aversion** ($\alpha{=}\beta{=}0.88, \lambda > 1$). The primary substrate is `NetworkManagement-v1` (Hubbs et al. 2020), a six-node multi-echelon inventory network (Oroojlooyjadid et al. 2022; Gijsbrechts et al. 2022; Leluc et al. 2023); transfer environments are MPE Cooperative Navigation (Lowe et al. 2017) and Level-Based Foraging (Christianos, Schäfer, and Albrecht 2020). We choose this over combat benchmarks such as SMAC (Samvelyan et al. 2019) deliberately: measuring how w moves the boundary needs a sequential social dilemma with a directed resource flow, whereas SMAC-style tasks couple spatial configuration and credit assignment with the quantity we isolate; we then confirm the *direction* on MPE and LBF to rule out a supply-chain artifact. Agents run PPO (Schulman et al. 2017) via Stable-Baselines3 (Raffin et al. 2021) and PettingZoo/SuperSuit (Terry et al. 2021; Terry, Black, and Hari 2020) (20 seeds/condition, 1,524 runs). Full setup is in the Supplement.

The Boundary and Its Behavioral Shift

A sharp, metastable boundary. Collapse is a sharp logistic transition in w, not a gradual decay: at 300k steps $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$ (transition width $\Delta w = 0.199$, $p = 3.2 \times 10^{-7}$), with collapse fractions rising from near-zero to near-one within $\Delta w < 0.3$ (Figure 1a). The boundary is not a constant but a *metastable* threshold that slides left with the training horizon, and across four matched risk-neutral horizons ($w_{\text{crit}} = 0.822, 0.666, 0.622, 0.573$ at $150\text{k}, 300\text{k}, 450\text{k}, 900\text{k}$) the slide is *log-linear in* T: $w_{\text{crit}} = 2.44 - 0.138 \ln T$ (RMSE 0.027; all four CIs of width $\leq$0.04; Figure 1b). This log-linear form is *consistent with* metastable escape (a Kramers-type first-passage picture; §): each doubling of the budget buys a fixed $\approx$ 0.10 leftward shift in w, so the boundary keeps falling below the longest measured point (0.573 at 900k), extrapolating the law to 2M steps gives $w_{\text{crit}} \approx 0.44$. Collapse at intermediate w is therefore the metastable *approach* to the deterministic floor $w^0_{\text{crit}} = c/(c+g)$ the mechanism derives (§), escapable with more training, not absorbing; we treat the floor's exact value as below the measured range rather than extrapolating the log-linear form to an (unphysical) zero crossing. This horizon dependence does not make w_{crit} arbitrary: a log-linear slide is the signature of a metastable, not ill-defined, threshold, at any fixed budget it is a well-defined number, and it quantifies a second design axis, the rate at which extra training rescues a sub-boundary policy (trading compute against centralization). The slide is not a supply-chain artifact either: under centralized-critic MAPPO it recurs in both transfer environments, on MPE w_{crit} falls $0.82 \to 0.68$ as the horizon grows 2M $\to$ 8M, and on Level-Based Foraging $0.33 \to 0.20$ over 1M $\to$ 4M, both with non-overlapping intervals. A basin-control experiment rules out initialization as the driver: within-w correlations between initial and converged base-stock are near zero (-0.11 to -0.38), so the collapse label does not track where agents start (Supplement).

Loss aversion raises the boundary; curvature lowers it. Against a single canonical null, $w^{\text{null}}_{\text{crit}} = 0.666$ $[0.635, 0.694]$, independently reproduced at 0.685 by a dedicated experiment (every other null-like number is a labeled offset, not a separate baseline), we sweep $\lambda \in \{1, \ldots, 7\}$ ($\alpha{=}\beta{=}0.88$, 300k). The boundary rises monotonically with loss aversion, $\{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\}$, then saturates near 0.85 with clear headroom to the $w{=}1$ ceiling, so the plateau is behavioral, not a ceiling artifact (Figure 2, Table 1); the endpoints are non-overlapping by 0.221. Curvature acts in the *opposite* direction, *lowering* w_{crit} to 0.586 (-0.100 $[-0.172, -0.033]$ vs. the null), that a single "risk-averse" psychology pulls the boundary in opposite directions is the first of two surprises, visible only against a calibrated null.

Two ways to count the loss-aversion effect. The magnitude depends on the baseline, and we report both (Table 1). Against the *linear null* (0.685) the combined CPT shift is +0.089 at $\lambda{=}3$ and +0.166 at $\lambda{=}7$, but this nets loss aversion against curvature's -0.100; the *pure loss-aversion increment* against the curvature baseline (0.586) (the correct isolate) is larger, +0.188 and +0.265 (reconciled by the offset, $-0.100 + 0.265 \approx 0.166$). Throughout, "the loss-aversion effect" means the increment over the curvature baseline; the net-vs-null figure is the deployment total. The full sweep is tracked by a one-parameter, theory-bounded affine law (§) to RMSE 0.014, and a finer $w \in \{0.80, 0.90\}$ grid tightens the steep $\lambda{\geq}5$ cells to CI widths $\approx$ 0.03 (half those from a single $w{=}0.85$ cell).

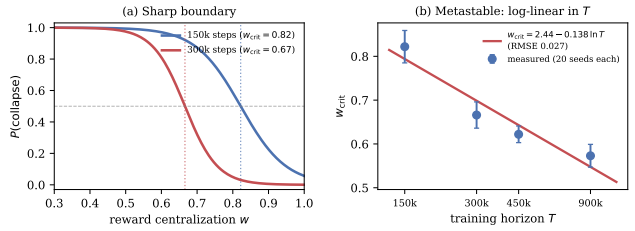


Figure 1: (a) Collapse is a sharp logistic transition in w; the boundary sharpens and slides left from 150k to 300k steps. (b) Across four matched risk-neutral horizons (20 seeds each, all CIs of width $\leq$0.04) w_{crit} falls *log-linearly* in T, $w_{\text{crit}} = 2.44 - 0.138 \ln T$ (RMSE 0.027), consistent with the metastable escape the mechanism predicts.

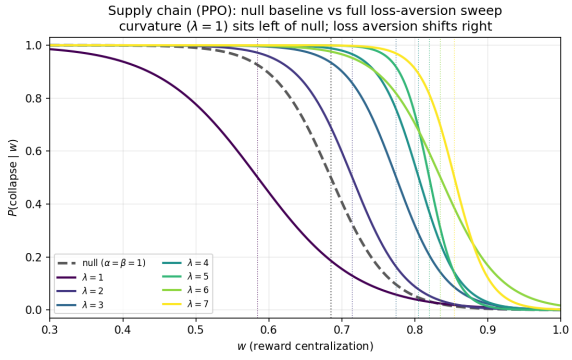


Figure 2: Behavioral decomposition and full loss-aversion sweep (supply chain, PPO). Curves are Firth-fitted $P(\text{collapse} \mid w)$; dotted lines mark w_{crit}. Curvature (λ=1) sits *left* of the null (easing coordination); loss aversion (λ=2, ..., 7) shifts w_{crit} *right*, rising then saturating.

A Mean-Field Mechanism Fixes the Sign

The measurement above is the paper's contribution; this section adds a *supporting* account of why the boundary exists and which way each component moves it. It is deliberately modest, sign and form, not magnitude, and not the basis of any empirical claim, so the results stand whether or not the reader accepts it. Following mean-field treatments of multi-agent learning (Barfuss, Donges, and Kurths 2019; Hu, Leung, and Leung 2019; Hussain, Belardinelli, and Leonardos 2025), we model each Gaussian-policy agent's mean action as a Langevin variable $\mathrm{d}\mu_i = D(w)\,\mathrm{d}t + \sqrt{2\Theta(w)}\,\mathrm{d}W$ (details in the Supplement), with drift D and a centralization-weighted diffusion Θ into which partner exploration enters. The replenishing regime makes an upstream node's inventory a local cost but a global benefit, giving cooperative drift $D(w) = \eta[wg - (1-w)c]$. Escape from the lazy basin is governed not by drift but by its ratio to gradient noise,

$$\mathrm{SNR}(w) = \frac{D(w)}{\sqrt{\Theta(w)}} \gtrsim \kappa_c, \tag{2}$$

and the crossing defines w_{crit}. Treating escape as a Kramers first-passage problem (Kramers 1940), the noise-induced analogue of a saddle-node bifurcation (Strogatz 1994), gives two expectations *consistent with* the data: collapse switches abruptly across w_{crit} and $w_{\text{crit}}(T)$ falls as T grows. It also makes a *falsifiable* prediction we test directly: raising exploration noise should raise w_{crit}, but only where that noise is the *binding* term. An entropy ablation confirms this as a *dissociation* (Figure 3): w_{crit} rises with the entropy coefficient on the supply chain and LBF (sparse rewards, injected entropy dominates) but stays flat on MPE (dense reward, high intrinsic exploration swamps the added term). The effect appears exactly where the mechanism predicts and is *absent* where it predicts none, a sharper test than uniform replication, and the one prediction a skeptic cannot dismiss as single-substrate luck. The mechanism does *not* deliver the value of w_{crit}: c, g, σ, κ_c are calibrated, and two pre-registered probes to localize it from the drift were inconclusive (survivor bias in the contested band; Supplement). The magnitude thus stays calibrated; the mechanism's content is the sign and the form.

Table 1: Three-way decomposition of w_{crit}. We report the effect both ways: net vs. the linear null (combines curvature and loss aversion), and the pure loss-aversion increment vs. the curvature baseline (0.586). "Sig." = difference CI excludes zero.

Condition	w_{crit}	vs. null	vs. curv.
Null (λ=1)	0.685	–	–
Curvature (λ=1)	0.586	$-0.100^{\checkmark}$	0
Loss av. λ=3	0.774	$+0.089^{\checkmark}$	+0.188
Loss av. λ=7	0.851	$+0.166^{\checkmark}$	+0.265

We are direct about the one place the account could be read as circular. The sign result below assumes a payoff premise ($\partial_\lambda F < 0$: loss aversion inflates loss-domain gradient variance, lowering the cooperative signal-to-noise ratio). We do *not* take this on faith or infer it from the outcome, we verify it independently with a finite-difference probe of the substrate (Supplement), so the proposition's content is the non-trivial claim that this *locally* measured sign *propagates* to the *global* boundary through the intractable noise, not the assumption that loss aversion hurts coordination.

Proposition 1 (Sign propagation of the loss-aversion shift)**.** *Let the order parameter x evolve as $\mathrm{d}x_t = F(x_t, w, \lambda)\,\mathrm{d}t + \sqrt{\Sigma(x_t, N)}\,\mathrm{d}W_t$ with $\Sigma > 0$, and* assume the payoff premise $\partial_\lambda F < 0$ *on the escape interval, i.e. loss aversion scales only the loss domain and so inflates gradient noise. Then $\partial_\lambda w_{\text{crit}}^{\text{Emp}} > 0$.*

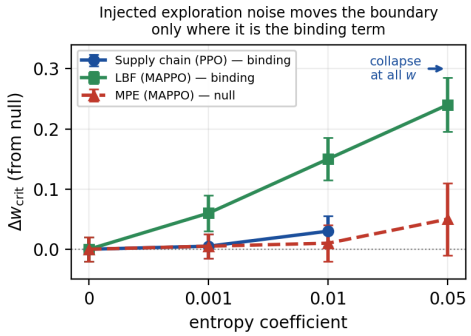


Figure 3: Entropy-ablation dissociation. Injected exploration noise raises w_{crit} on the supply chain (PPO) and LBF (MAPPO), where it is the binding noise term, but leaves the MPE boundary flat (overlapping CIs), where dense-reward intrinsic exploration dominates, a prediction that correctly stays null is evidence of mechanism, not curve fitting.

Proof sketch. The Kramers barrier $\Delta U = \int 2F/\Sigma \, \mathrm{d}z$ has $\partial_\lambda \Delta U < 0$ pointwise, since $\Sigma > 0$ and $\partial_\lambda F < 0$ throughout, independent of the sign of F; a lower barrier shortens escape ($\partial_\lambda \tau < 0$) while $\partial_w F > 0$ lengthens it. The implicit function theorem on $\tau(w, \lambda) = T$ gives $\mathrm{d}w_{\text{crit}}^{\text{Emp}}/\mathrm{d}\lambda = -\partial_\lambda \tau / \partial_w \tau > 0$. The magnitude depends on the intractable Σ; the direction does not. The same argument with curvature's drift signs reversed yields the opposite shift. □

We state this as a *proposition*, not a theorem, because its force is conditional: it assumes the payoff premise $\partial_\lambda F < 0$ rather than deriving it from first principles. We verify that premise directly on the substrate (a finite-difference probe confirms loss aversion inflates the loss-domain gradient variance; Supplement), so the result's content is that this *local* sign propagates to the *global* boundary regardless of the intractable noise Σ, which is exactly what makes the direction (but not the magnitude) predictable. The same drift analysis fixes the opposite sign of the two prospect-theory components without fitting: curvature ($\alpha, \beta < 1$) compresses both gains and losses, lowering the variance in Θ, raising the SNR, and so *lowering* w_{crit}, while loss aversion inflates Θ and *raises* it, matching the measured -0.100 and $+0.265$ (Table 1).

A theory-bounded law, tested out of sample. The same loss-domain split fixes the functional *form*, not just the sign. Only the loss domain scales with λ, giving

$$w_{\text{crit}}(\lambda) = \frac{\phi(\lambda)}{\phi(\lambda) + g/c}, \qquad \phi(\lambda) = 1 + \rho(\lambda - 1), \quad (3)$$

with $\rho \in [0, 1]$ *bounded by the theory*, a genuine parametric constraint, not a free curve fit. We fix g/c from the λ=1 baseline and $\rho = 0.61$ from three calibration points ($\lambda \in \{1, 3, 7\}$), then test *out of sample*

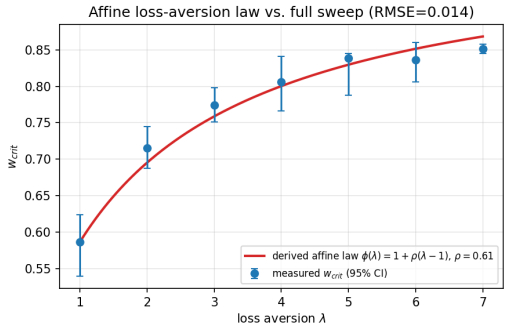


Figure 4: The derived affine law (3) (line, one free parameter $\rho = 0.61$ fixed on three points) against the measured $w_{\text{crit}}(\lambda)$ (points, bootstrap 95% CIs). It tracks the held-out points to RMSE 0.014, with a mild concave residual at the high-λ plateau.

Table 2: What the mechanism claims. The predictive content is sign and form; the magnitude is calibrated, and we make no closed-form claim for w_{crit}.

Derived (no fit)	opposite signs; monotone $w_{\text{crit}}(\lambda)$; affine ϕ with $\rho \in [0, 1]$; abrupt sharpness; w_{crit} falls with T
Confirmed	opposite signs, affine form; horizon slide (also MPE, LBF); exploration-noise dependence
Calibrated	g/c; loss-domain share ρ; noise scale
Not claimed	closed-form w_{crit}; magnitude on any axis

against the held-out four ($\lambda \in \{2, 4, 5, 6\}$): $w_{\text{crit}}^{\text{pred}} = \{0.586, 0.695, 0.759, 0.800, 0.830, 0.851, 0.868\}$ versus $w_{\text{crit}}^{\text{meas}} = \{0.586, 0.715, 0.774, 0.806, 0.839, 0.836, 0.851\}$, RMSE 0.014 across all seven points (Figure 4). The fit does not depend on which points are held out: under leave-one-out cross-validation (fit ρ on six points, predict the seventh, repeated for each) ρ stays in $[0.59, 0.64]$ (mean 0.61, s.d. 0.02) and the held-out RMSE is 0.016, so the law is not an artifact of the chosen calibration split. The one systematic residual is a mild concave overshoot at the plateau, consistent with agents endogenously reducing loss exposure as λ grows. We report this derived, bounded law rather than a better-fitting power law because the point is mechanism, not goodness of fit, and we are explicit that it is *single-substrate*: it does not transfer to LBF (§). Table 2 states what the mechanism derives a priori, what is empirically confirmed, and what stays calibrated.

The Instrument Travels; So Does the Direction

The first thing that must generalize is the *instrument*: the pre-registered boundary is only useful if it can be

measured in environments beyond the supply chain. It can, we estimate a CI'd w_{crit} in two further environments and under a second algorithm, and using it there, we ask whether the behavioral finding travels. The mechanism commits in advance to a direction (loss aversion inflates gradient noise, raising w_{crit}); if that is not an artifact of one substrate or optimizer, the sign should survive changing both. We match the learner to each environment (independent PPO where coordination is locally learnable; centralized-critic MAPPO (Yu et al. 2022) where reward is shared-global). The direction is robust in every setting; the absolute level, curvature sign, and functional form are not (Figure 5, Table 4). On the *supply chain under MAPPO* (only the learner changes), w_{crit} rises 0.796 $\rightarrow$ 0.889, monotone from λ=2, converging with PPO by $\lambda \geq 4$. On *MPE* the loss-aversion excess is +0.158 at λ=7 while curvature is null (+0.042, CI spans zero); on *LBF* curvature moves the boundary *right*, and the affine law misfits it by RMSE 0.15, both confirming the curvature sign and form are substrate-specific. Thus the *method*, a CI'd boundary that loss aversion shifts, generalizes across four environment-algorithm combinations, while the affine constants and curvature sign do not.

Two boundary conditions, both predicted. The mechanism makes the boundary contingent on two ingredients, and ablating each confirms it (Table 3). *Action representation.* The signal-to-noise crossing is a property of a continuous gradient, so destroying it should remove the boundary. Value-based learners (QMIX, IQL) on a discretized K=11 action space give a flat, w-independent collapse rate, but a single-variable control isolates the cause: PPO on the *same* discrete shell also goes flat, so it is coarse discretization, not the value-based paradigm, that erases the boundary. This is exactly the continuous-gradient prediction: coarse bins snap the policy to fixed attractor levels, suppressing the gradient noise near the boundary so agents never visit the metastable lazy region where collapse is decided, a scope condition of the mechanism, not a fragility of the method (and not a claim that discreteness per se removes the boundary: MPE is natively discrete and retains it). It also scopes whether the loss-aversion direction holds under value-based methods: since those learners show no measurable boundary here, the question is not well-posed and we claim nothing about it. *Credit assignment.* On a shared-reward task the boundary is measurable only with a learner that can coordinate at all: independent PPO collapses at *every* w on MPE (a coordination floor), while centralized-critic MAPPO recovers it. Each is a confirmed scope condition, not a negative result.

A Minority Sets the Boundary

The decomposition treats the population as homogeneous, but deployed systems mix loss-averse and risk-neutral participants. Holding the substrate, PPO config, label, and grid identical to the main sweep, we ask two questions: how much bias at a critical node is needed to move the boundary, and which node's bias matters most (Figure 6).

Table 3: Action-representation ablation. The control (last row) changes only the action space while holding the learner at PPO, isolating coarse discretization as what erases the boundary.

Learner	Action space	Boundary?
PPO / MAPPO	continuous	yes
QMIX (centralized value)	discrete K=11	no
IQL (independent value)	discrete K=11	no
PPO (control)	discrete K=11	**no**

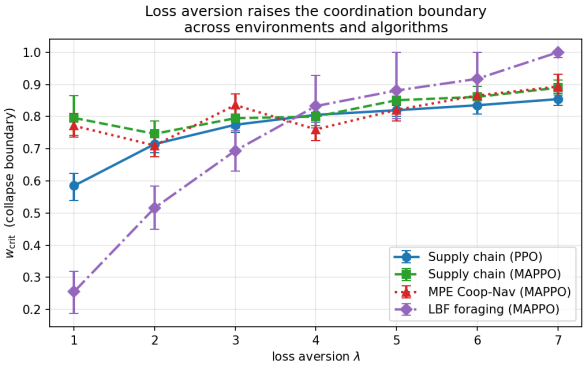


Figure 5: w_{crit} versus λ across four settings: supply chain under PPO and MAPPO, MPE under MAPPO, LBF under MAPPO (bootstrap 95% CIs). w_{crit} rises with λ in every setting; the *direction* generalizes, while levels (most at λ=1), curvature sign, and exact form are substrate-specific. LBF $\lambda \geq 5$ points extrapolate past the w=1 ceiling (outside the measured grid) and are labelled exploratory.

A single node, dose for dose. Sweeping the most downstream node's $\lambda \in \{1, \ldots, 7\}$ (others risk-neutral) traces a strictly monotone dose-response: w_{crit} rises 0.586 $\rightarrow$ 0.856, converging on the *fully* loss-averse value (0.851) by λ=7 (Figure 6b). One node, biased hard enough, imposes the entire network's requirement, the homogeneous result is the saturation of a continuous law, not a special case.

Position: proximity to demand, not chain index. Placing one λ=7 node at each position, the effect is strongly position-dependent, but the governing variable is how directly a node's output reaches demand, not a chain index (Figure 6a). The two nodes feeding retail directly dominate (0.856, 0.800); the upstream supplier feeding *only* the dominant downstream node is nearly as strong (0.797), while those *splitting* output across both are weaker (0.726, 0.673). The network is a branching DAG, not a chain, and w_{crit} tracks *path concentration*: a biased node matters more the more directly and ex-

Table 4: Generalization of the loss-aversion *direction*. The sign holds everywhere; only the supply chain (PPO) yields the affine law. Detail in the Supplement.

Setting (algo.)	Loss-av. effect	Affine law?
Supply (PPO)	+0.166 (λ=7)	yes (0.014)
Supply (MAPPO)	0.80→0.89	–
MPE (MAPPO)	+0.158 (λ=7)	no
LBF (MAPPO)	0.26→0.83	no (0.15)

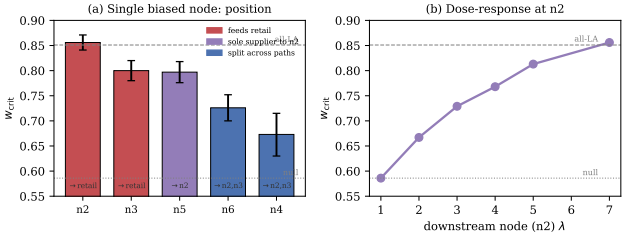


Figure 6: A minority sets the boundary. (a) A single λ=7 node by position: nodes feeding retail directly (red) dominate; among upstream suppliers, the one feeding *only* the dominant downstream node (purple) nearly matches them, while those *splitting* output across paths (blue) are weaker, w_{crit} tracks path concentration to demand, not chain index. (b) Dose-response: sweeping the downstream node's λ traces a strictly monotone rise to the homogeneous all-loss-averse value (0.851, dashed).

clusively its product flows to demand, so audit the participant whose output most directly reaches demand, not the average or merely "upstream" one (cf. the EGT tipping-point result of Mengesha, Sun, and Roy 2025, here in deep-MARL with a topological signature). The result is single-substrate *by construction*: "which node sets the boundary" is well-posed only under a directed resource flow, which MPE and LBF lack. We therefore report the cascade as a *falsifiable prediction, not a tested generalization*, testing it needs a second directed-flow substrate, and is future work, not an oversight.

Discussion

That probability weighting and loss aversion exert opposite effects on the boundary is, to our knowledge, novel; prior work applies prospect theory as a unified model without decomposing it (Lalmohammed et al. 2025; Ghaemi et al. 2024). The operational reading is concrete: a system of λ=7 agents needs $w > 0.851$ for reliable coordination against $w > 0.685$ for risk-neutral agents, a +0.166 deployment margin that scales with λ, so domains nearer $\lambda \approx 2$–3 incur a smaller but positive cost. Our phase-diagram framing sits within a recognized line of multiagent phase-transition results (Yamaguchi 2025; Capannoli et al. 2026) and the broader cooperative-AI project (Dafoe et al. 2020; Hammond et al. 2025; Barfuss et al. 2020); its distinctive contribution is a boundary set by the policy-gradient signal-to-noise ratio rather than a fixed update rule.

What a practitioner does with this. The boundary is a design quantity: a team can estimate its own w_{crit}, budget for the loss-aversion margin rather than discover collapse in production, and (where w is capped) treat λ (set by the training objective (Ethayarajh et al. 2024; Rabin 2000)) as a second lever. The minority result adds a targeting rule: audit the biased participant whose output most directly reaches demand, not the average agent. This targeting is dual-use, the same finding that lets a defender *harden* the most influential participant equally identifies the node whose induced bias would most efficiently degrade coordination, which is why we report the decomposition and cascade openly, so the effect stays auditable rather than hidden.

Scope and Future Work. Three boundaries each point to a concrete next step. *(i) From one substrate to many.* The quantitative core (the $w_{\text{crit}}(\lambda)$ law and the cascade) is single-substrate by construction: the direction, boundary, and metastable slide replicate across environments and a second algorithm, but the magnitudes belong to one simulator and the affine law does not transfer to LBF. Replicating the cascade on a second directed-flow domain would make it a tested generalization rather than a prediction. *(ii) From calibrated to predictive.* The mechanism fixes sign and form but not magnitude; localizing w_{crit} from the learning dynamics resisted two pre-registered probes, both of which track *where the policy sits*, not *where the boundary is* (Supplement). A localizing observable is the clearest open problem the paper leaves. *(iii) From bracket to calibration.* Estimating the effective λ of preference-aligned agents from behavior would tie the behavioral axis to the training pipeline. None undercuts the central, replicated claim, each marks where measurement ends and prediction begins.

Conclusion

We presented a confidence-intervalled diagram of cooperative-MARL coordination collapse, and used it to show, a priori and across four environment-algorithm combinations, that loss aversion raises the centralization a team needs to coordinate. Beyond that finding, the pre-registered protocol is a reusable instrument: any cooperative-MARL study can report *where* its system collapses, with error bars, rather than asserting that centralization helps. Crucially, it is *agnostic to the mechanism inducing collapse*: prospect-theoretic bias is only its first application, and the same measurement applies to reward design, communication, coordination incentives, or other algorithms, a quantitative alternative to the qualitative descriptions of coordination collapse the field currently relies on.

Ethical Statement

This work uses a simulated substrate with no human subjects or personal data. The dual-use reading of the minority result and the governance implications of raising centralization are discussed above; substrate-scoped numbers should be re-measured before deployment.

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